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NEONATAL HYPOTHERMIA AND NEONATAL HYPOGLYCEMIA

Temperature Regulation

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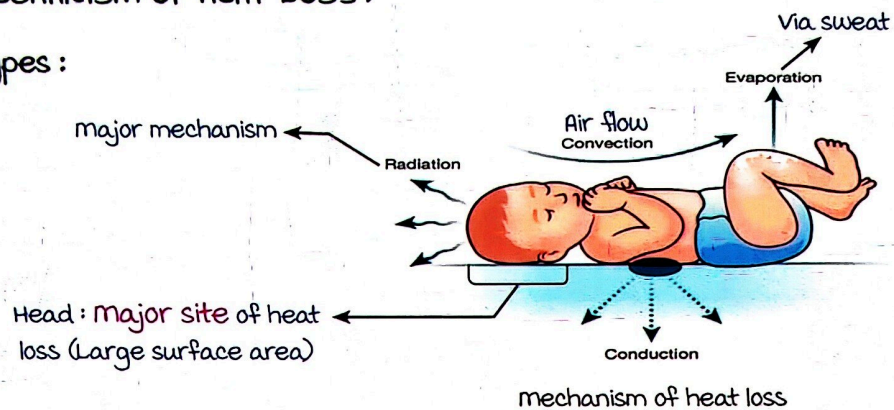
Normal temperature : $36.5 - 37.5^{\circ}\text{C}$.

Site : Roof of dry axilla.

Instrument : Electronic thermometer.

MECHANISM OF HEAT LOSS :

Types :



SOURCES OF HEAT PRODUCTION :

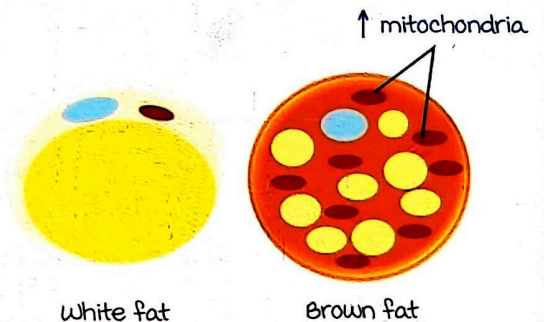
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1. ↑ physical activity.
2. **Brown fat** : Non shivering thermogenesis in early months of life.

• Locations :

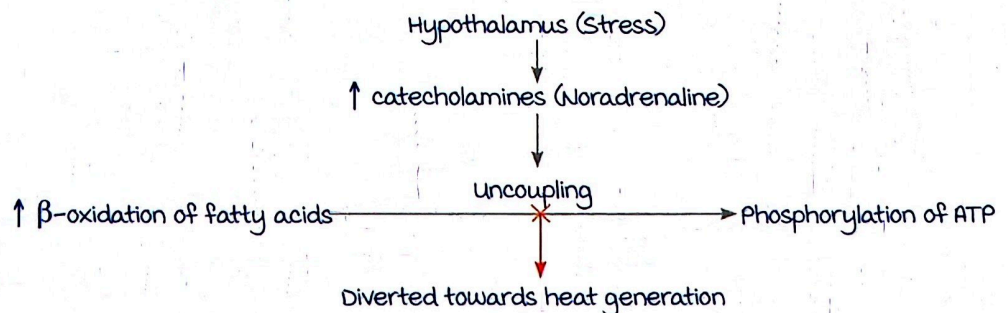
- **Nape of neck** (m/c).
- Interscapular region.
- Peri renal region.

- mechanism : **uncoupling of oxidative phosphorylation**.



White fat

Brown fat

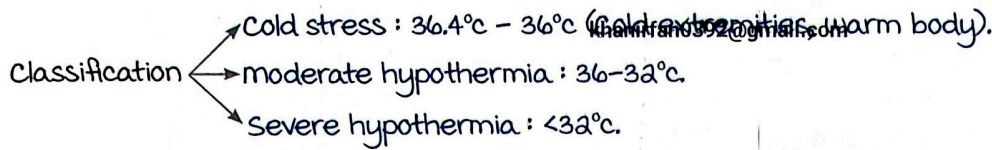


Neonatal Hypothermia & Neonatal Hypoglycemia

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HYPOTHERMIA :

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Note : Hyperthermia $> 37.5^{\circ}\text{C}$.

PREVENTION OF HYPOTHERMIA :

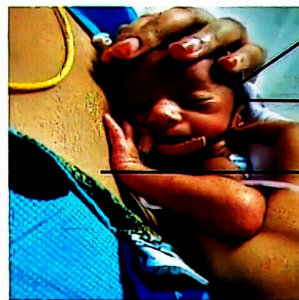
Temperature maintenance is a critical aspect of newborn care.

In a hemodynamically stable baby :

1. Cover the body from head to toe.

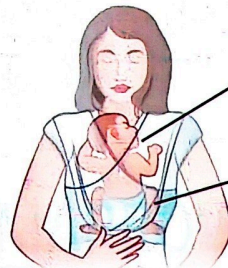
2. Kangaroo mother care (KMC) :

- In **LBW** or **preterm babies**.
- Nutrition : Breastfeed.
- Duration : As long as possible.
- Mother can take naps during KMC in semi recumbent position (30° elevation).
- Avoid < 1 hour sessions : Repeated handling of the baby $\rightarrow$ Stress.
- Stop KMC once baby attains ≥ 2.5 kg or at 37 weeks.



KMC

- Turn baby's head to one side
- Baby kept close to mother
- Skin to skin contact : Heat transfer by conduction



Position of baby in KMC

In a hemodynamically unstable baby :

Placed in NICU.

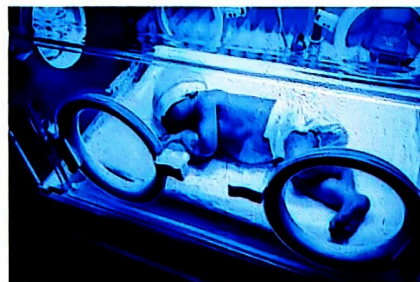
1. Overhead radiant warmer : Open source.

- Heat transfer by radiation.
- Operating modes
 - $\rightarrow$ manual.
 - $\rightarrow$ Auto/servo : Preferred.



Overhead radiant warmer

2. Incubator : Closed/covered source of heat.



Incubator

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Neonatal HypoglycemiaBlood glucose <45 mg/dL.**SCREENING :**

Hypoglycemia can be asymptomatic and still have dangerous consequences



Important to screen, detect & treat.

method : Heel prick test.

Indication : Routine new born care.

Risk factors :

Inadequate substrate	Relative hyperinsulinemia	Sick neonates (NICU)
- Small for gestational age (SGA) : $<3^{\text{rd}}$ percentile. - Preterm : <35 weeks.	- Large for gestational age. - Infants of diabetic mother (IDM).	- Sepsis. - Birth asphyxia. - Severe hypothermia.

Duration : First 72 hours of life $\rightarrow$ High risk period.**MANIFESTATIONS :**

1. Asymptomatic

2. Symptomatic

• Jitteriness/tremors :

- m/c & earliest sign.
- Stops with a stimulus.

• Seizure : Not stimulus sensitive.

• Lethargy.

• Refusal of feeds.

• Autonomic symptoms (uncommon).

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MANAGEMENT :

Symptomatic :

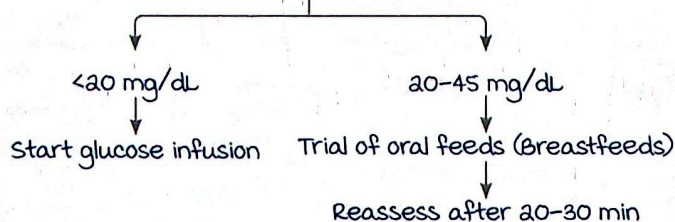
IV dextrose 2 mL/kg of 10% dextrose



Continuous glucose infusion

Glucose infusion rate (GIR) : 6 mg/kg/min $\xrightarrow{\text{2 mg/kg/min increments}}$ 12 mg/kg/min (Depending on persistence of hypoglycaemia)

Asymptomatic : Based on glucose level



RESISTANT HYPOGLYCEMIA

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Definition : GIR requirements exceed $>12 \text{ mg/kg/min}$ or hypoglycemia persists for $>7 \text{ days}$ after starting treatment.

major causes :

- Hypopituitarism : D/t $\downarrow$ ACTH.
- Metabolic disorders : Glycogen storage disorder, galactosemia.
- Nesidioblastosis :
 β - Islet cell hyperplasia of pancreas $\rightarrow$ Congenital hypoglycaemia.

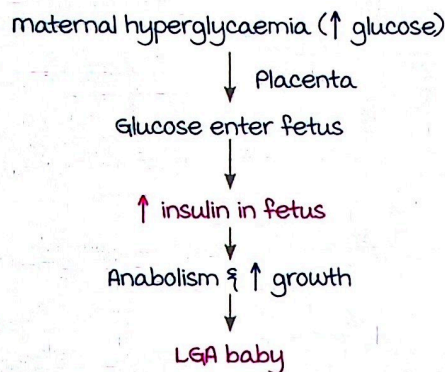
management :

1. Stimulate gluconeogenesis : Hydrocortisone, glucagon.
2. $\downarrow$ insulin production : Diazoxide, octreotide.

Note : Do not use diazoxide and glucagon in SGA babies.

INFANT OF DIABETIC MOTHER (IDM)

Pedersen's hypothesis :



Effects of $\uparrow$ insulin :

1. Asymmetric septal hypertrophy : Resolves after birth (when insulin $\downarrow$).
2. Inactivation of surfactant : Alveoli collapse $\rightarrow$ Respiratory distress syndrome (RDS).
3. Lazy left colon syndrome : $\downarrow$ gut motility $\rightarrow$ Delayed passage of meconium.

Side effects in LGA baby :

$\uparrow \text{O}_2$ requirement $\rightarrow$ $\uparrow$ RBC production/polycythemia $\rightarrow$ Hyperviscosity $\rightarrow$ Thromboembolism

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Neonatology

Congenital malformations :

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Etiopathology :

mothers with overt uncontrolled diabetes

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Hyperglycaemia in 1st trimester (Embryogenesis)↓
Congenital anomaly.

Defects seen :

Cardiac defects → m/c : VSD.
 → most specific : Transposition of great arteries.

CNS defects → m/c : Neural tube defects (NTD).
 → most specific : Caudal regression syndrome (Lumbosacral agenesis)
 Poorly developed lower limbs.

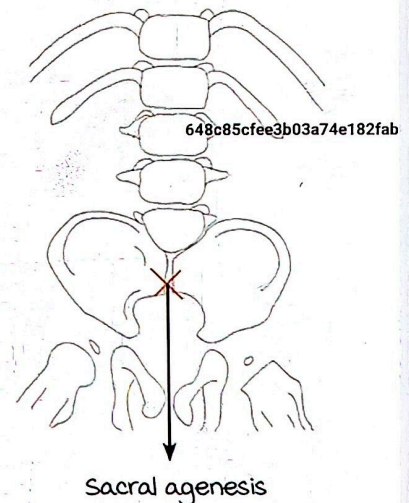


Caudal regression syndrome

metabolic abnormalities :

1. Hypocalcemia & hypomagnesemia :

↑ insulin

↓
Interferes with PTH release↓
Functional hypoparathyroidism↓
Hypocalcemia & hypomagnesemia(Ca²⁺ < 8 mg/dL) (Mg²⁺ < 1.5 mg/dL)

Sacral agenesis

2. Hypoglycemia :

m/c cause of seizure (Hypoglycemia > hypocalcemia).